

From Autoregressive to Diffusion Across Model Families: Parameter-Efficient Conversion of Instruction-Tuned Language Models

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Abstract

Masked diffusion language models offer flexible generation orders and parallel token updates, but competitive systems are commonly pretrained from scratch at substantial computational cost. We study whether diffusion generation can instead be added to existing instruction-tuned autoregressive language models through a single parameter-efficient conversion procedure. Our method replaces causal attention with bidirectional attention and trains low-rank adapters under the likelihood-based masked-diffusion objective used by LLaDA, while keeping the pretrained backbone, token embeddings, and language-model head frozen. We apply the same procedure to three model families of comparable scale: Llama 3 8B, Qwen2.5 7B, and Gemma 2 9B. Each converted model is evaluated against its autoregressive parent and against LLaDA 8B Instruct, allowing us to measure both capability retention and incremental conversion cost. [TODO: Complete with the main quantitative findings after the experiments.]

1 Introduction

Autoregressive language models generate text by factorizing a sequence from left to right. This factorization has enabled effective scaling and strong instruction-following models, but it fixes the order in which tokens are generated and makes each new token depend on the preceding decoding steps [Vaswani et al., 2017]. Masked diffusion language models (MDLMs) provide an alternative: they begin from a corrupted sequence and iteratively predict multiple masked positions using bidirectional context [Austin et al., 2021, Sahoo et al., 2024, Shi et al., 2024]. This formulation supports flexible generation orders, iterative refinement, and a tunable trade-off between the number of denoising steps and generation quality.

The strongest evidence that masked diffusion can scale to general-purpose language modeling comes from LLaDA [Nie et al., 2025]. LLaDA was pretrained from scratch on 2.3 trillion tokens and subsequently supervised fine-tuned on 4.5 million instruction–response pairs. It demonstrated in-context learning, instruction following, and competitive performance on general, mathematical, and coding benchmarks. Its training recipe also has a probabilistic interpretation: response tokens are independently masked at a randomly sampled noise level, and the model minimizes a weighted cross-entropy objective over masked positions. However, LLaDA 8B required approximately 130,000 H800 GPU-hours for pretraining. Repeating this process for every desired model size, architecture, or domain remains impractical for most research groups.

An appealing alternative is to reuse representations already learned by autoregressive models. DiffuLLaMA showed that autoregressive checkpoints can be converted into diffusion language models through continued training [Gong et al., 2025]. LAD subsequently explored a more parameter-efficient route, using low-rank adaptation (LoRA) and bidirectional attention to convert Llama models using instruction data [Hu et al., 2022, Kuiper et al., 2025]. LAD demonstrated that such models can generate and iteratively revise complete responses after a comparatively small amount of training. Nevertheless, the evidence remains incomplete in three respects. First, LAD was evaluated only on one architecture family, making it unclear whether its conversion procedure is specific to Llama. Second, its combined masking and structural-corruption objective differs from the likelihood-based MDLM objective used by LLaDA, complicating direct comparison. Third, prior evaluations do not cleanly separate the quality of a converted model from the capabilities already present in its autoregressive parent.

We therefore ask:

Can a single parameter-efficient training recipe convert instruction-tuned autoregressive models from different architecture families into useful masked diffusion language models, and how much of each parent model’s capability is retained?

We introduce LAD-Generic, a model-agnostic conversion procedure that combines bidirectional attention, LoRA adapters, and the LLaDA masked-diffusion supervised fine-tuning objective. We apply the procedure without family-specific algorithmic changes to Llama 3 8B Instruct, Qwen2.5 7B Instruct, and Gemma 2 9B IT. To avoid conflating conversion quality with differences between pretrained models, each diffusion model is paired with its unchanged autoregressive parent under a common evaluation protocol. LLaDA 8B Instruct provides an external diffusion baseline. We account for efficiency as the incremental cost of conversion—explicitly treating autoregressive pretraining and instruction tuning as prerequisites rather than claiming that the complete lifecycle is cheaper.

Our contributions are:

- a unified, parameter-efficient recipe for converting instruction-tuned autoregressive models to masked diffusion generation;
- the first [TODO: verify after the final literature search] controlled cross-family study of LoRA-based diffusion conversion across Llama, Qwen, and Gemma models of comparable scale;
- a paired evaluation that quantifies capability retention relative to each autoregressive parent, alongside comparison with a purpose-trained diffusion model; and
- an analysis of incremental training tokens, GPU-hours, trainable parameters, memory use, and the quality–speed trade-off at inference.

2 Method

2.1 Problem formulation

Let p denote an instruction prompt and $r_0 = (r_0^1, \dots, r_0^L)$ its clean response. We start from an instruction-tuned autoregressive language model with parameters ϕ . Our goal is to learn a conditional masked-diffusion model $p_{\phi, \theta}(r_0 | p)$ while updating only a small subset θ of the model parameters. The resulting model must generate a response from a fully masked sequence and refine it through parallel token updates.

We define conversion as an incremental post-training stage. The compute used to pretrain and align the parent checkpoint is therefore not attributed to the conversion algorithm. We report this upstream training, where publicly known, separately from conversion tokens, GPU-hours, and trainable parameters.

2.2 Backbone models

We study three publicly available instruction-tuned decoder-only model families with similar parameter counts:

- Llama 3 8B Instruct;
- Qwen2.5 7B Instruct; and
- Gemma 2 9B IT.

These checkpoints were also represented among the autoregressive comparisons in the LLaDA study, reducing differences in scale and model generation. For each parent model, we construct one diffusion-adapted counterpart. The original checkpoint remains unchanged and serves as a paired autoregressive baseline. LLaDA 8B Instruct is evaluated as a separately pretrained diffusion baseline.

2.3 Architectural conversion

Standard decoder-only transformers prevent a token from attending to future positions. During conversion, we remove this causal restriction for the response span, allowing each response position to attend bidirectionally to the prompt and the complete corrupted response. Prompt tokens remain fully observed and are never prediction targets. Padding positions are excluded from attention and loss computation.

We insert LoRA adapters into the query, value, and output projections of every self-attention layer. A single rank q and scaling rule are used across all three families. All normalization parameters are trainable. Transformer weights, token embeddings, and the language-model head remain frozen. This choice tests whether the pretrained vocabulary projection can be reused rather than relearned during diffusion conversion.

We report LoRA parameters, unfrozen dense parameters, and total trainable parameters separately. Before full training, we require that the models differ by no more than three percentage points in trainable-parameter fraction and by no more than 100 million trainable parameters in absolute count. If a fixed rank violates either condition, we use the nearest feasible rank and disclose the deviation. [TODO: Insert the selected rank and exact counts after the parameter audit.]

Mask representation. The corruption process uses a single mask token. [TODO: Decide whether to reuse an existing single-token representation or add a dedicated token. If a new token is added while embeddings remain frozen, train only its input-embedding row and report it separately.] The mask token is used only as corrupted input and is never a clean prediction target.

2.4 Masked-diffusion objective

We follow the supervised fine-tuning objective of LLaDA [Nie et al., 2025]. For every prompt–response pair, we draw a noise level $t \sim \mathcal{U}(0, 1]$. Each response token is independently replaced with [MASK] with probability t , yielding r_t . The prompt is left uncorrupted. The loss is

$$\mathcal{L}(\theta) = -\mathbb{E}_{t,p,r_0,r_t} \left[\frac{1}{tL} \sum_{i=1}^L \mathbf{1}[r_t^i = \text{[MASK]}] \log p_{\phi,\theta}(r_0^i | p, r_t) \right]. \quad (1)$$

Only corrupted response positions contribute to the loss. The $1/t$ weighting matches the likelihood-based masked-diffusion objective and places the conversion experiments on the same probabilistic footing as LLaDA. Unlike the original LAD formulation, we do not apply token swaps, duplications, span shifts, or an additional loss on visible response tokens.

2.5 Instruction data and sequence construction

The initial conversion corpus reuses the public mixture constructed for LAD [Kuiper et al., 2025]. It contains approximately 951,000 instruction–response pairs drawn from general instruction following, multiple-choice reasoning, mathematics, and code. [TODO: List dataset versions, licenses, filtering rules, mixture weights, deduplication, and exact post-filter counts.]

Examples are formatted using each parent model’s native chat template. Only response tokens are corrupted and scored. Mini-batches use dynamic sequence lengths up to 1,024 tokens and are bucketed by length to reduce padding. End-of-sequence tokens are treated as ordinary response tokens: they may be masked and contribute to Equation 1. This teaches the model to terminate responses during diffusion sampling. Examples exceeding the context limit are [TODO: specify truncation or exclusion policy].

We begin with this fixed corpus for all architectures. We expand the mixture only if pilot experiments reveal a shared, pre-specified failure across model families, such as poor mathematical reasoning, code generation, or response length control. Any expansion is applied identically to every converted model and documented as a protocol revision.

2.6 Optimization

All converted models use the same optimizer family, learning-rate schedule, LoRA dropout, and effective token batch size. Training proceeds for a fixed token budget rather than a fixed number of padded sequences. We save checkpoints at common token counts, allowing conversion efficiency and convergence to be compared across architectures.

[TODO: Specify LoRA rank and scaling, AdamW parameters, peak learning rate, warmup, decay, precision, gradient clipping, effective token batch size, checkpoint interval, random seeds, number of runs, and stopping rule.]

2.7 Diffusion sampling

Given a prompt p and target response length L , generation starts from a fully masked response r_1 . We discretize the reverse process into N uniformly spaced timesteps. At every step, the model predicts all currently masked positions in parallel. Tokens are provisionally filled using [TODO: greedy decoding or the selected sampling rule]; the least-confident predictions are then remasked so that the number of visible tokens follows the next timestep. This low-confidence remasking strategy follows LLaDA and is used for the main results.

Generation terminates at the first predicted end-of-sequence token, and tokens after it are discarded. We evaluate multiple values of N to characterize the quality–latency trade-off, but select one decoding configuration on a validation set before evaluating test benchmarks. [TODO: Specify generation lengths, step grid, temperature, tie handling, EOS-confidence treatment, and validation selection rule.]

2.8 Evaluation design

The primary analysis measures *capability retention*: the performance of each converted model relative to its own autoregressive parent under matched prompts and task scoring. We report absolute differences and normalized ratios, while avoiding ratios for metrics whose zero point lacks a ratio-scale interpretation. The second analysis compares the converted models with LLaDA 8B Instruct under the closest reproducible decoding and prompting protocol.

Following LLaDA, the benchmark suite covers general knowledge and reasoning (MMLU, MMLU-Pro, HellaSwag, and ARC-Challenge), mathematics and science (GSM8K, MATH, and GPQA), and code generation (HumanEval and MBPP). We use deterministic, task-specific scoring whenever a reference answer or executable test is available. LLM-as-a-judge evaluation is reserved for open-ended instruction-following samples and is reported separately from the benchmark results. [TODO: Specify benchmark versions, splits, few-shot settings, prompt templates, answer extraction, pass@ k , judge model, judge rubric, and uncertainty intervals.]

We additionally report repetition statistics, response-length distributions, conditional diffusion loss on held-out responses, throughput, latency, peak accelerator memory, and performance as a function of denoising steps. Perplexity from an unrelated autoregressive evaluator is treated only as a secondary fluency diagnostic because it is not directly comparable to either autoregressive likelihood or the diffusion likelihood bound.

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